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20 / 20 submitted

Q1. Union of Sets

Score: 0.00 TC: 0/1 passed

CODE

```
A = {1, 2, 3}
B = {3, 4, 5}

print(A | B)
```

OUTPUT

{1, 2, 3, 4, 5}

Q2. Intersection

Score: 0.00 TC: 0/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

C = A.intersection(B)
print(C)
```

OUTPUT

{2, 3}

Q3. Difference

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

print(A - B)
```

INPUT

{1,2,3}
{2,3,4}

OUTPUT

{1}

Q4. Symmetric Difference

Score: 0.00 TC: 0/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

result = A.symmetric_difference(B)
print(result)
```

OUTPUT

{1, 4}

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Q5. Subset

Score: 5.00 TC: 1/1 passed

CODE

```
A = {2, 3}
B = {1, 2, 3, 4}

if A.issubset(B):
    print("Subset")
else:
    print("Not Subset")
```

OUTPUT

Subset

Q6. Superset

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1,2,3,4}
B = {2,3}

if A.issuperset(B):
    print("Superset")
else:
    print("Not Superset")
```

OUTPUT

Superset

Q7. Disjoint Sets

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2}
B = {3, 4}

if A.isdisjoint(B):
    print("Disjoint")
else:
    print("Not Disjoint")
```

OUTPUT

Disjoint

Q8. Remove Duplicates

Score: 0.00 TC: 0/1 passed

CODE

```
list1 = [1,2,2,3,3,4]
result = set(list1)
print(result)
```

INPUT

1 2 2 3 3 4

OUTPUT

{1, 2, 3, 4}

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Q9. Common Elements

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}
C = {2, 5}

common = A & B & C
print(common)
```

OUTPUT

{2}

Q10. Missing Numbers

Score: 0.00 TC: 0/1 passed

CODE

```
given = {1, 2, 4, 5, 7, 10}
all_nums = set(range(1, 11))

missing = all_nums - given
print(missing)
```

OUTPUT

{8, 9, 3, 6}

Q11. First Repeated

Score: 5.00 TC: 1/1 passed

CODE

```
arr = list(map(int, input().split()))

seen = set()
for x in arr:
    if x in seen:
        print(x)
        break
    seen.add(x)
```

INPUT

10 20 30 20 40

OUTPUT

20

Q12. Duplicate Elements

Score: 0.00 TC: 0/1 passed

CODE

```
arr = list(map(int, input().split()))

seen = set()
dupes = set()

for x in arr:
    if x in seen:
        dupes.add(x)
    else:
        seen.add(x)

print(dupes)
```

INPUT

10 20 30 20 10

OUTPUT

{10, 20}

Q13. Compare Lists

Score: 0.00 TC: 0/1 passed

CODE

```
list1 = list(map(int, input().split()))
list2 = list(map(int, input().split()))

print(set(list1) == set(list2))
```

INPUT

```
1 2 2 3
3 2 1
```

OUTPUT

True

Q14. Duplicate Tuples

Score: 0.00 TC: 0/1 passed

CODE

```
tuples_list = [(1,2),(1,2),(3,4)]

result = list(set(tuples_list))
result.sort()

print(result)
```

INPUT

[(1,2),(1,2),(3,4)]

OUTPUT

[(1, 2), (3, 4)]

Q15. Pair Sum

Score: 0.00 TC: 0/1 passed

CODE

```
arr = list(map(int, input("List: ").split()))
target = int(input("Target: "))

pairs = []
n = len(arr)

for i in range(n):
    for j in range(i + 1, n):
        if arr[i] + arr[j] == target:
            pairs.append((arr[i], arr[j]))

print(pairs)
```

INPUT

```
2 4 6 8 10
12
```

OUTPUT

List: Target: [(2, 10), (4, 8)]

Q16. Unique Characters

Score: 5.00 TC: 1/1 passed

CODE

```
s = input()
print(len(set(s)))
```

INPUT

programming

OUTPUT

8

Q17. Unique Words

Score: 0.00 TC: 0/1 passed

CODE

```
sentence = input().strip()
words = sentence.split()
unique_words = set(words)
print(unique_words)
```

INPUT

```
python is easy python
{'python', 'is', 'easy'}
```

OUTPUT

```
{'is', 'easy', 'python'}
```

Q18. Exactly One Set

Score: 0.00 TC: 0/1 passed

CODE

```
A = {1, 2, 3}
B = {3, 4, 5}
C = {5, 6, 7}

result = (A - (B | C)) | (B - (A | C)) | (C - (A | B))
print(result)
```

OUTPUT

```
{1, 2, 4, 6, 7}
```

Q19. Menu Driven**CODE**

```
# Menu Driven Set Operations

a = list(map(int, input("Enter the first set elements: ").split()))
b = list(map(int, input("Enter the second set elements: ").split()))

A = set(a)
B = set(b)

while True:
    print("\nMenu")
    print("1. Union")
    print("2. Intersection")
    print("3. Difference (A - B)")
    print("4. Symmetric Difference")
    print("5. Exit")

    choice = input("Enter your choice: ").strip()

    if choice == '1':
        print("Union:", A.union(B))
    elif choice == '2':
        print("Intersection:", A.intersection(B))
    elif choice == '3':
        print("Difference:", A.difference(B))
    elif choice == '4':
        print("Symmetric Difference:", A.symmetric_difference(B))
    elif choice == '5':
        print("Program terminated.")
        break
    else:
        print("Invalid choice. Please enter a number between 1 and 5.")
```

OUTPUT

```
Traceback (most recent call last):
  File "C:\Windows\TEMP\sandbox_192420031RD_6a71f836d2d799.83618234\main.py", line 3, in <module>
    a = list(map(int, input("Enter the first set elements: ").split()))
                                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
EOFError: EOF when reading a line
```

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Q20. Student Database**CODE**

```
def display_menu():
    print("\nEnter your choice:")
    print("1. Add Student")
    print("2. Remove Student")
    print("3. Search Student")
    print("4. Display Students")
    print("5. Count Students")
    print("6. Exit")

def main():
    students = set()

    while True:
        display_menu()
        choice = int(input())

        if choice == 1:
            name = input("Enter student name: ").strip()
            students.add(name)
            print("Student added successfully.")

        elif choice == 2:
            name = input("Enter student name to remove: ").strip()
            if name in students:
                students.remove(name)
                print("Student removed successfully.")
            else:
                print("Student not found.")

        elif choice == 3:
            name = input("Enter student name to search: ").strip()
            if name in students:
                print("Student found.")
            else:
                print("Student not found.")

        elif choice == 4:
            print(f"Students: {students}")

        elif choice == 5:
            print(f"Total number of students: {len(students)}")

        elif choice == 6:
            print("Program terminated.")
            break

        else:
            print("Invalid choice. Please try again.")

if __name__ == "__main__":
    main()
```

OUTPUT